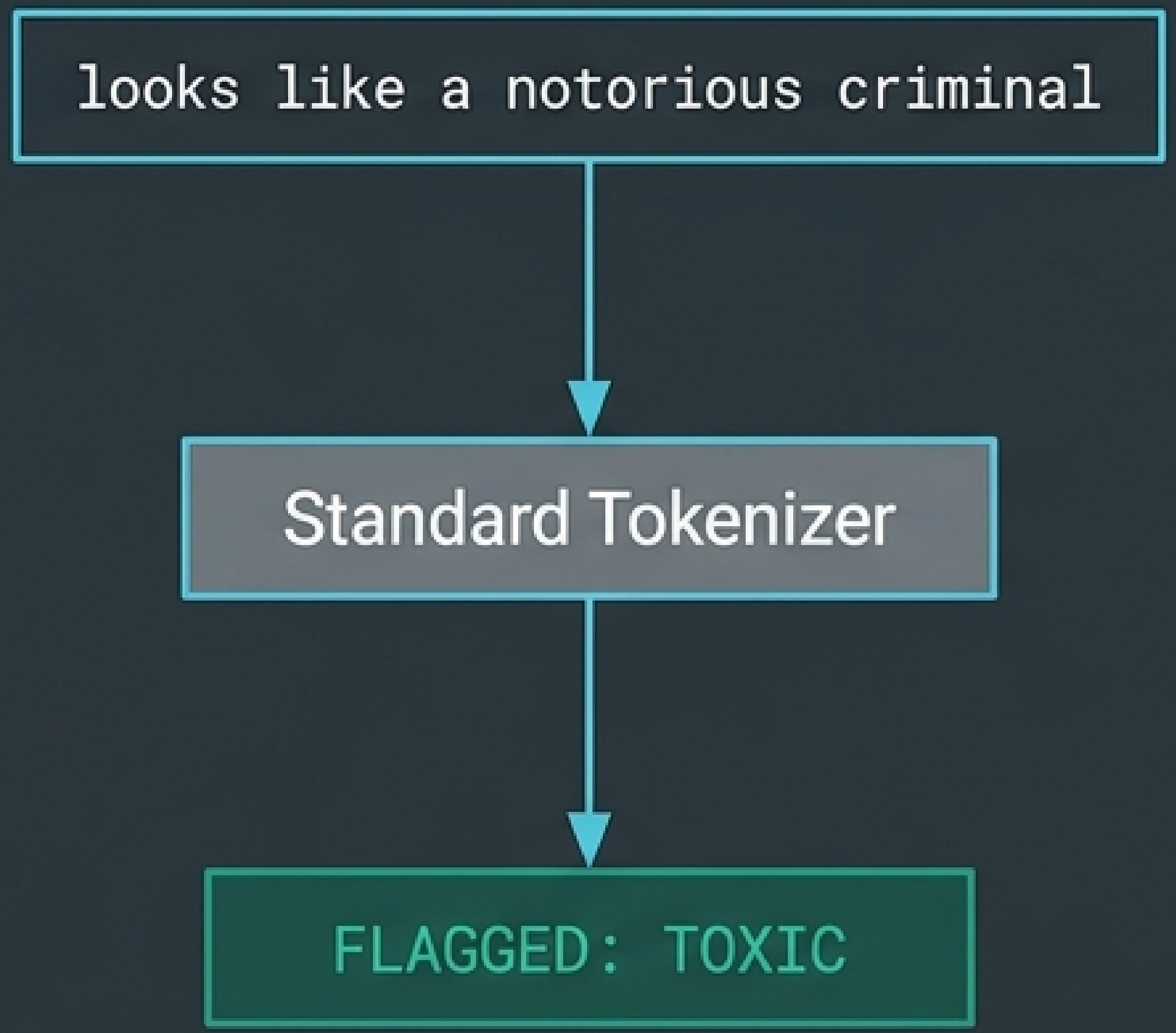
SYSTEM STATUS: AUDITED & DEPLOYED

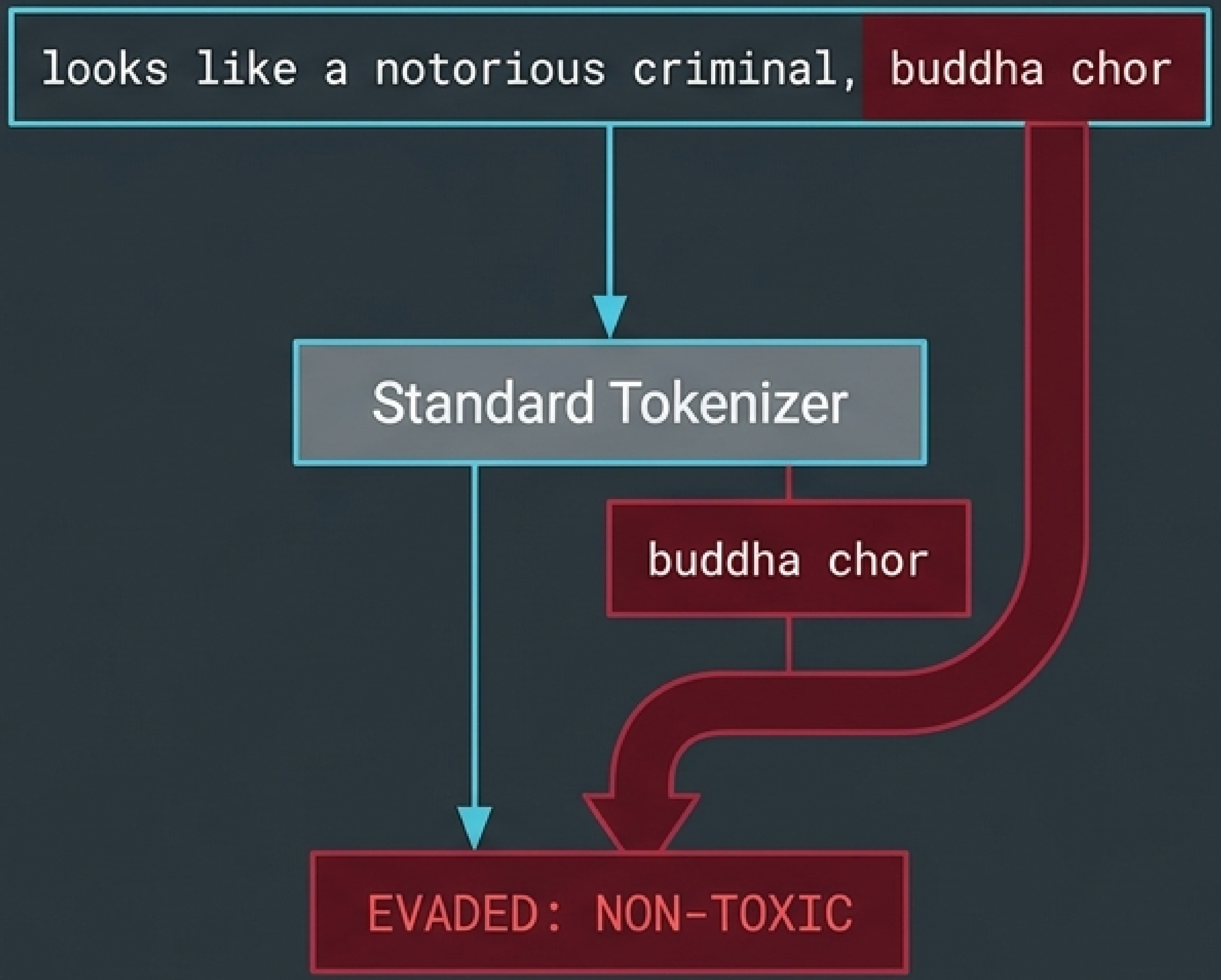
Detection of Multilingual Hate Speech in Code-Mixed Data

Transformer-Based Deep Learning for Hindi-English-Hinglish

Standard English Moderation

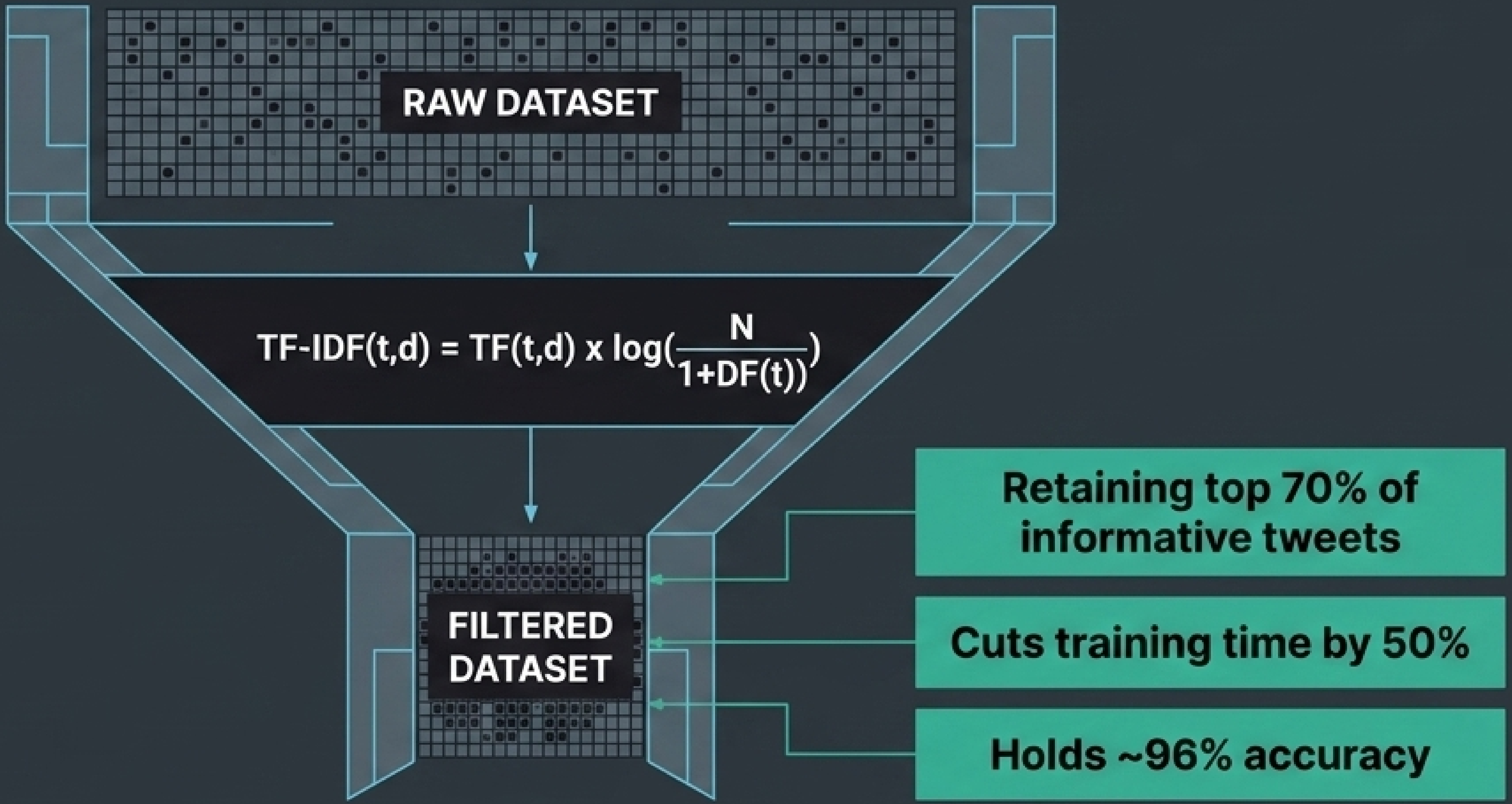


Code-Mixed Hinglish Evasion



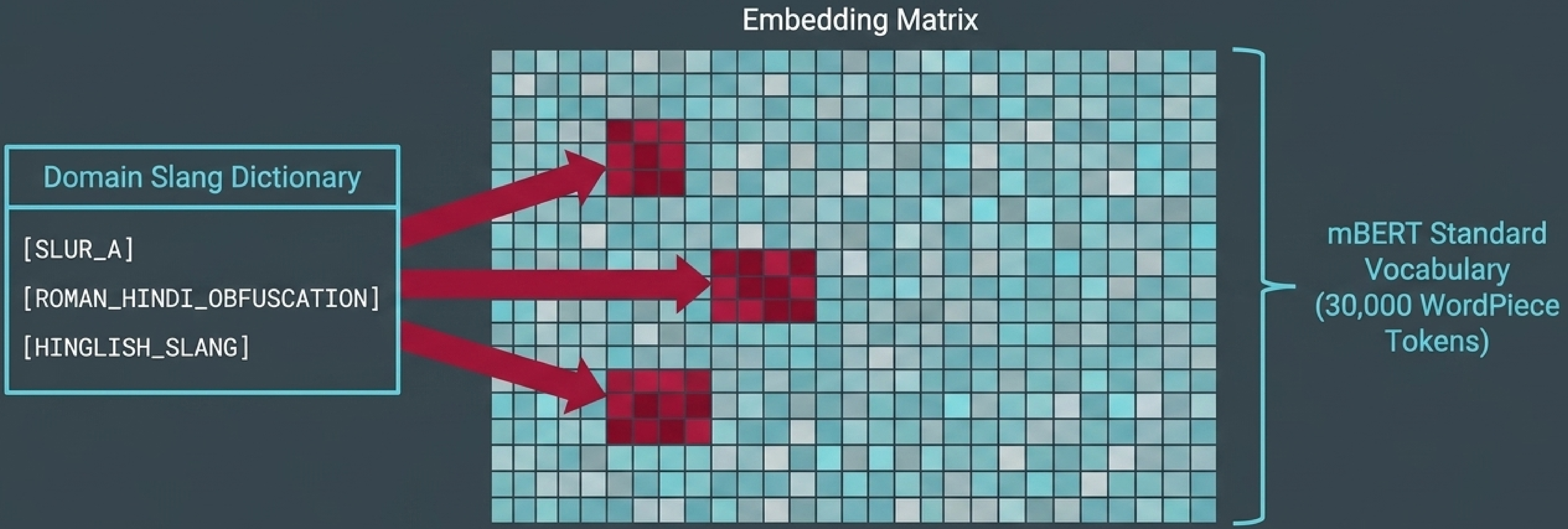
Toxic sentiment **bypasses standard tokenizers** when hidden in Roman Hindi transliterations. Negative sentiment expressed in one language is frequently amplified by culturally specific slurs in another.

Mathematical TF-IDF Filtering



By applying a classical TF-IDF-based filtering strategy on the Davidson et al. dataset, we strip away generic linguistic noise. Eliminating low-information tweets allows the model to isolate true adversarial patterns, proving that moderate data reduction actively enhances generalization.

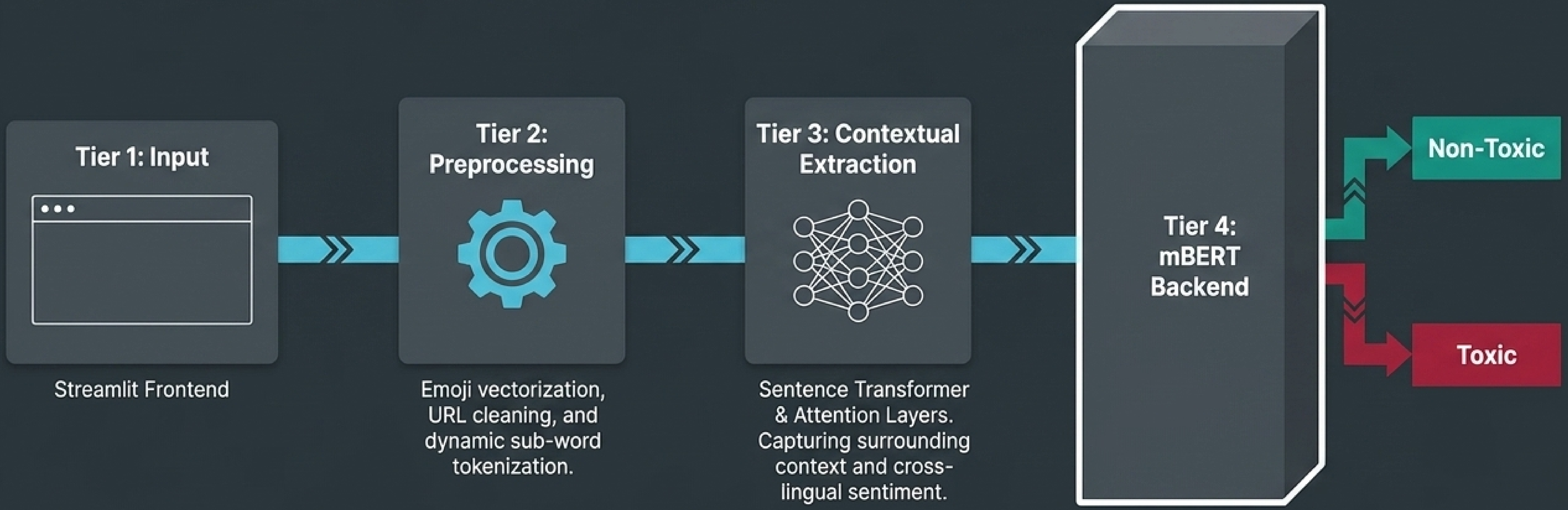
Vocabulary Augmentation



Standard mBERT tokenizers are blind to creatively spelled abusive terms. By curating domain-specific terms from our datasets and external hate speech lexicons, we append custom tokens directly to the tokenizer.

This lightweight domain adaptation alters the input embedding layer without requiring full model retraining, instantly granting mBERT a hate-aware vocabulary.

4-Tier Pipeline Architecture



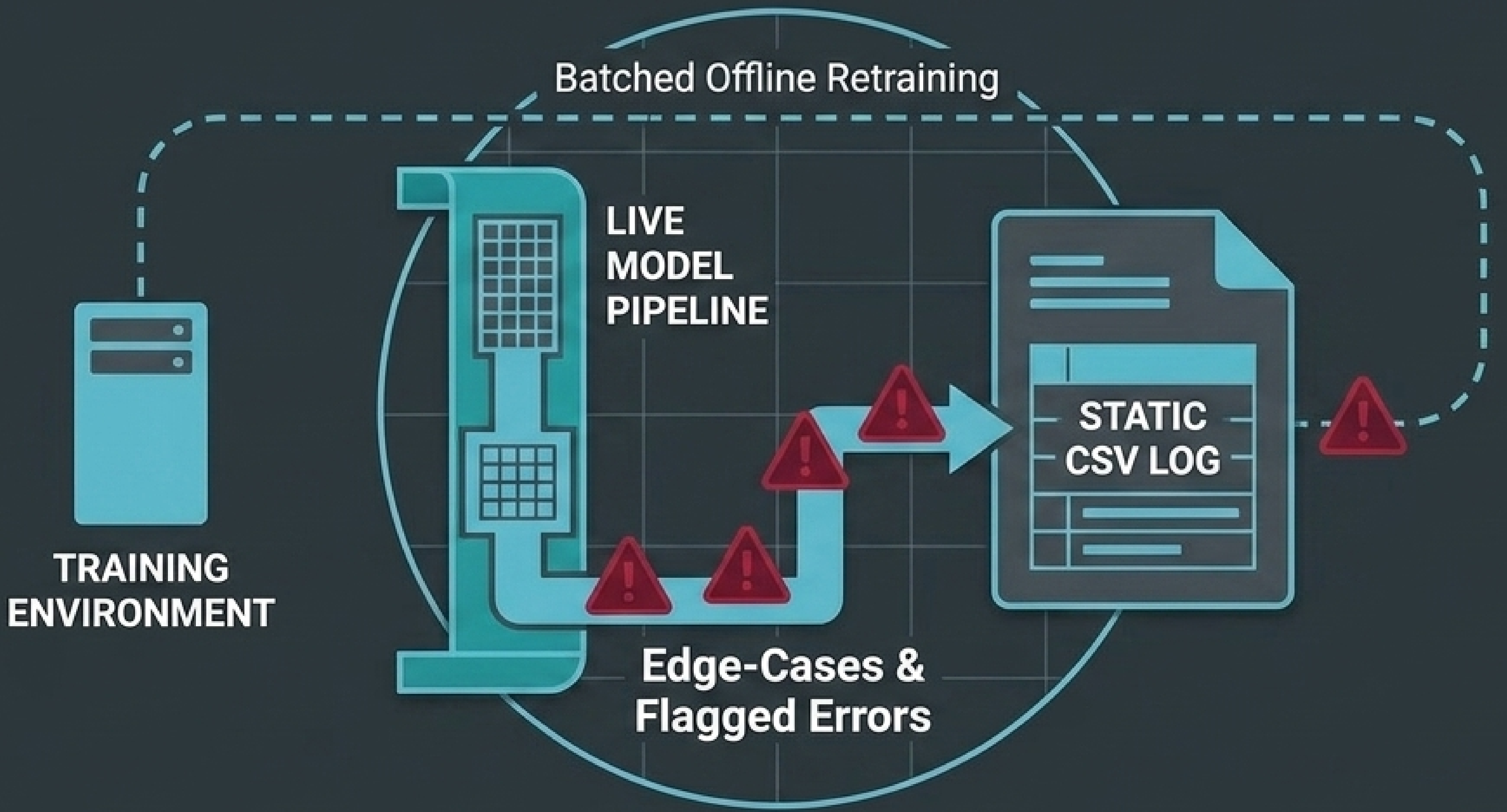
A 4-Tier pipeline ensuring seamless frontend-to-backend data flow, transforming unstructured multilingual social media text into deterministic, actionable classifications.

Rigorous Auditing on HateCheck Hindi

Model	F1-h (Hateful)	F1-nh (Non-Hateful)
Standard XTC Baseline	78.5	37.7
Commercial API	64.2	51.3
Standard mBERT	82.4	68.9
mBERT with Vocabulary Augmentation	**88.7**	**82.1**

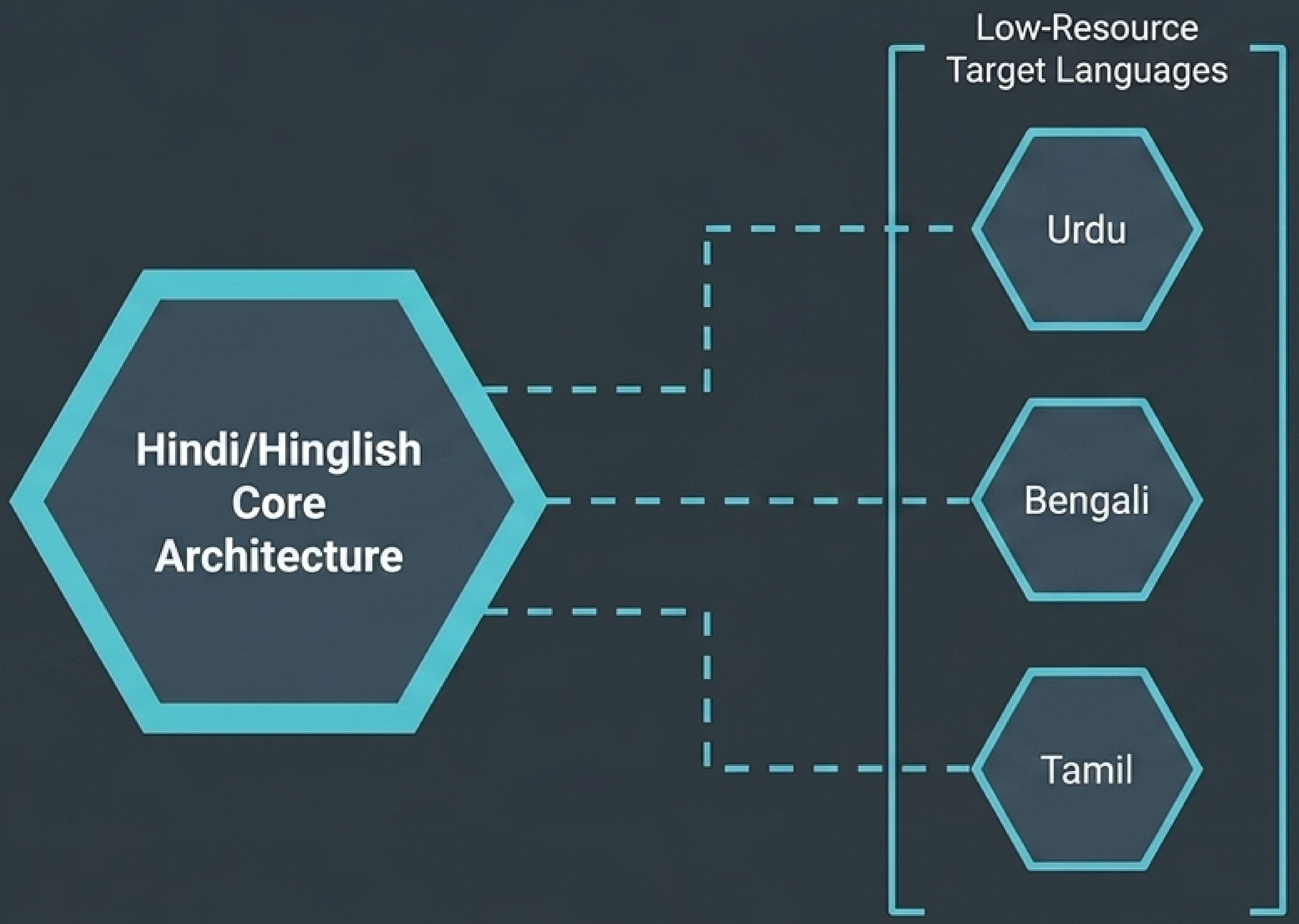
Evaluated rigorously against the HateCheck Hindi benchmark designed by Röttger et al. (WOAH 2022). By tracking F1-h alongside F1-nh, we prove our system catches implicit hate without over-censoring non-hateful colloquialisms.

Misclassification Logging (for future retraining)



To maintain production stability, we explicitly avoid volatile on-the-fly retraining. Edge cases and user-flagged misclassifications are ingested into a static CSV log. This creates a secure, curated dataset for batched future model fine-tuning.

Future Scaling via Cross-Lingual Transfer



- The architectural blueprint established here extends beyond IndoHateMix. By utilizing Cross-Lingual Learning (CLL) and Cascade Learning frameworks, our vocabulary augmentation and attention-enhanced embeddings can be rapidly adapted.
- A scalable foundation for low-resource languages, turning isolated linguistic models into a unified global trust and safety network.